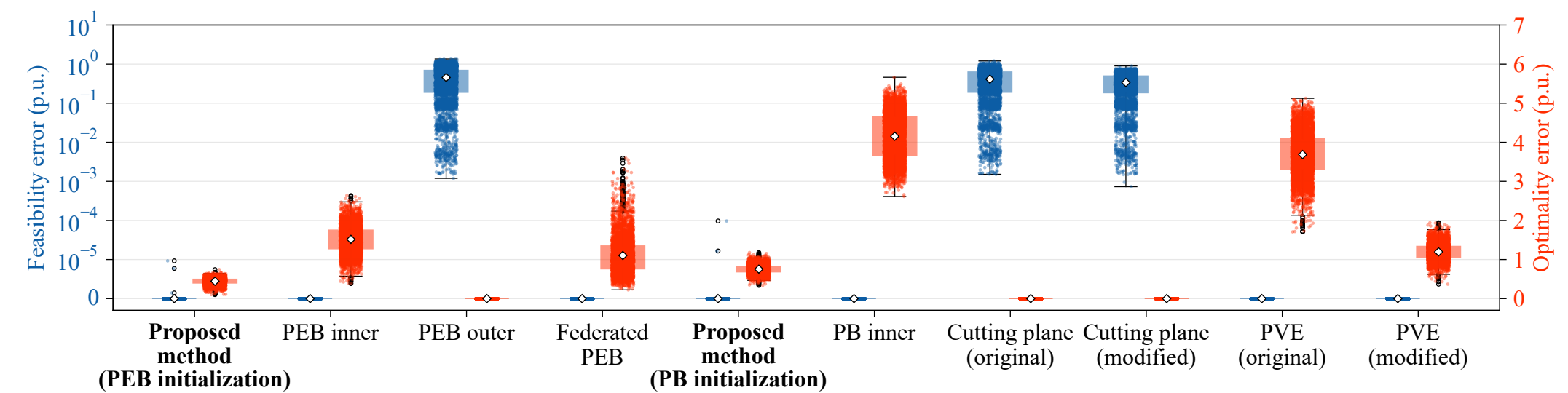


Evaluation directions per method: n = 5000. Points: all samples.
Box: interquartile range; diamond: mean; whiskers: 1.5 x IQR.
Feasibility < 1e-6: display floor only; statistics and mean markers use original values.

Federated PEB: Non-unique affine LP optima and redundant PEB constraints make the original full KKT Jacobians singular at initialization.
Numerical treatment: merge duplicate PEB rows (96 to 94); minimize off-diagonal L1 at trace = max(0, p - 1e-8 - 1e-9*abs(p)), where p is the maximum trace.
Selected-simplex-basis sensitivities replace direct KKT inversion; adaptive backtracking is used and device-containment constraints are retained.
This numerical variant does not certify a unique classical gradient or eliminate optimization stalling.